Titanic survival prediction

Research AI Task 7 | Logistic regression from scratch

Cross-validation selected **enhanced features**, an L2 penalty of **1** and a classification threshold of **0.5**. On 179 reserved passengers, accuracy was **83.80%**, F1 was **0.7752** and ROC-AUC was **0.8866**.

Data and preprocessing

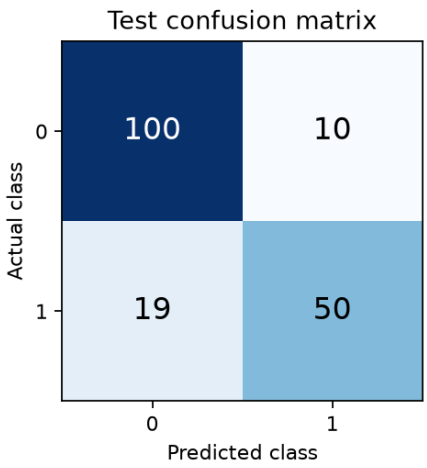
The supplied titanic.csv contains 891 passengers: 549 did not survive and 342 survived. There are 177 missing ages, 687 missing cabin values and two missing embarkation values. A stratified split with seed 42 reserved 179 passengers for testing and 712 for training.

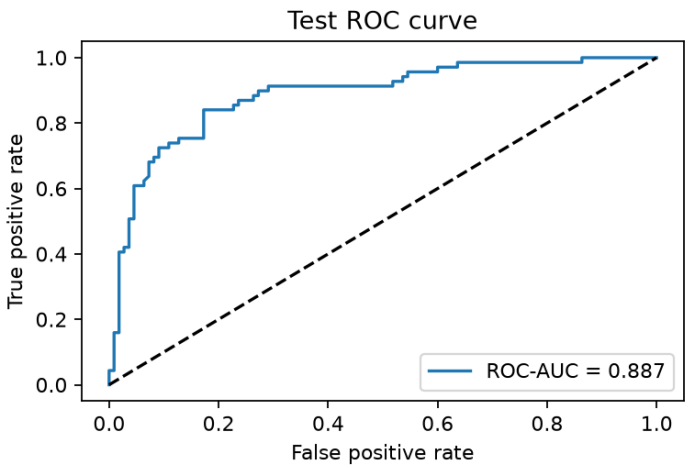
Training medians and modes replace missing values. One-hot categories and scaling are learned within each training fold. Passenger IDs, raw names, tickets and cabin strings are excluded. Enhanced predictors add passenger title, family size, travelling-alone status, cabin presence, missing-age status, child status, log fare and a sex/class interaction.

Evaluation metrics

Metric	Training	Cross-validation	Test
Accuracy (%)	84.97	83.33	83.80
Precision	0.8517	0.8247	0.8333
Recall	0.7363	0.7179	0.7246
F1	0.7898	0.7676	0.7752
Balanced accuracy (%)	82.83	81.15	81.69
ROC-AUC	0.8815	0.8589	0.8866
Log loss	0.3892	0.4225	0.4183

Survival is the positive class. Accuracy and balanced accuracy are percentages; precision, recall, F1 and ROC-AUC are fractions. Larger classification scores and smaller log loss are better. Class 0 means did not survive; class 1 means survived.





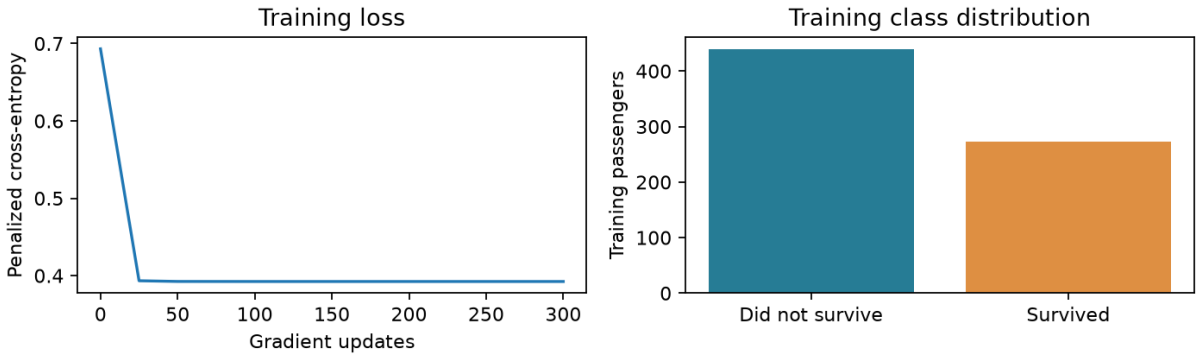
Model selection and verification

Highest validation accuracy

Features	Penalty	Threshold	CV accuracy	CV log loss
Enhanced	1	0.5	83.33%	0.4225
Enhanced	10	0.5	83.05%	0.4209
Enhanced	0.1	0.5	82.77%	0.4262

Five stratified folds were repeated three times. Eighteen feature/penalty/threshold combinations were ranked by pooled validation accuracy, then lower log loss and fewer encoded features. All imputation and scaling were refitted per fold. The test set was excluded from model and threshold selection.

All 90 validation fits converged. The selected model used 23 encoded predictors and converged in 300 full-batch gradient updates, with maximum absolute gradient 1.85e-08. The class implements fit, predict_proba and predict using NumPy. Momentum accelerates the updates; the intercept is unpenalized. Stable sigmoid and cross-entropy calculations avoid overflow.



Reserved test comparison

Model	Accuracy	F1	ROC-AUC
Logistic regression	83.80%	0.7752	0.8866
Majority-class baseline	61.45%	0.0000	0.5000

The model correctly classified 150 of 179 passengers, compared with 110 for the majority baseline. It missed 19 survivors and predicted survival for 10 non-survivors. The baseline always predicts non-survival; its probability estimate is the training survival rate.

Verification and limits

Checks covered numerical gradients, stable probabilities, manual metrics and tied ROC scores, disjoint stratified splits, missing inputs and unseen categories. Fresh-kernel executions reproduced the selected settings and results.

Fold accuracy standard deviation was 3.40 percentage points. Repeated folds share passengers; this is not a confidence interval. The historical sample and shared travel groups limit generalization.

Project repository